

Amiri Prescod

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WEBSITE: <https://amiriprescod.vercel.app/> | GITHUB: <https://github.com/aprescod12>

EDUCATION

Villanova University – College of Engineering, Villanova, PA

May 2026

Bachelor of Science in Electrical Engineering | Minor in Computer Science

GPA: 3.21

NCAA DI Athlete: Track and Field (60m & 100m BIG EAST Champion)

Courses: Embedded Systems, Signal Processing, Digital Systems, Microcontrollers and Applic, Med Device Tech

ENGINEERING & PROFESSIONAL EXPERIENCE

Independent Undergraduate Biomedical Research, Villanova University

Fall 2025-Spring 2026

Undergraduate Research Assistant

- Acquired and processed biomedical signals to analyze human physiological and neurological behavior.
- Troubleshoot and restored functionality of legacy sensor, improving data reliability for ongoing research.
- Developing a user-facing application to visualize and communicate real-time biomedical data.

The National Gas Company of Trinidad and Tobago Limited, Point Lisas, TTO

Summer 2025

Undergraduate Intern – Asset Maintenance

- Supported Electrical, Telecommunications, and Instrumentation teams in asset maintenance operations.
- Conducted site inspections, system troubleshooting, and operational diagnostics across industrial facilities.

Heft IQ, Remote

Summer 2024

Intern

- Applied foundational knowledge of AI and machine learning to evaluate real-world use cases.
- Analyzed potential AI applications and translated concepts into practical implementations through guided projects

SKILLS

Programming: Python, C, C++, MATLAB, JavaScript, TypeScript, VHDL, SQL

Interpersonal: Leadership, Teamwork, Critical Thinking, Time Management, Project Management, Communication

PROJECTS

Medical Monitoring & Safety Device | <https://github.com/aprescod12/medical-monitoring-safety-device>

- Developed an embedded health monitoring system integrating fall detection, sleep-state analysis, and hardware-triggered emergency alerts.
- Designed real-time sensor processing and deterministic event handling for rapid safety response.

Track and Field Training App | <https://github.com/aprescod12/track-training-app>

- Built a full-stack React Native app with Supabase to centralize workout logging and performance tracking
- Designed scalable database schema and optimized queries for history, calendar workouts, and performance data
- Used NCAA Division I athlete experience to iterate features based on real user feedback via TestFlight

ESP32 4-Floor Elevator Simulator | <https://github.com/aprescod12/esp32-elevator-simulator>

- Engineered a 4-floor elevator controller using SCAN scheduling and state-driven floor servicing logic.
- Optimized responsiveness through debounced inputs and cooperative multitasking.

Particle Photon Security System | <https://github.com/aprescod12/particle-photon-security-system>

- Developed a cloud-connected burglar alarm with motion detection and remote arming via Particle Cloud.
- Integrated signal filtering and time-based validation to reduce false positives in real-world conditions

LEADERSHIP & ACTIVITIES

NCAA D1 Student Athlete

October 2022-Present

Villanova University Track and Field (Dedicate 20 hr/week)

- NCAA D1 Sprinter; Team Leader, SAAC Representative, and International Student-Athlete Leader

NSBE

August 2023-Present

NovaNSBE Executive Board Member (Pre-College Initiative Chairperson and Senator)

- Organized outreach events engaging local high school students in STEM education.
- Represented NSBE Chapter at conferences and voted for the National and Regional NSBE Executive Positions.